

ARTEMII KARAPETIAN

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I design rendering and graphics systems and the structure behind interactive canvases — a GLSL fractal engine, a formally verified algebra of layered images. I write Python, and I specify and direct the hard core from first principles, implementing in other languages through agentic coding.

PROJECTS

Make Your Fractal — browser fractal explorer | *TypeScript, WebGL2/GLSL — agentic build* 2026

- Specified and directed the build of a fractal explorer: a small language (lexer, parser, AST) in which the user defines their own recursion, escape rule and free constant, transpiled to **GLSL** and evaluated per pixel so edits re-render live.
- Diagnosed the binding constraint — float64 loses precision at deep zoom, so raising precision globally cannot work — and specified the fix: track each pixel as a small delta from one high-precision reference orbit, differentiated automatically via dual numbers, verified against float64 ground truth (relative error 1.5×10^{-8}). Built via agentic coding. github.com/rtmkrptn/make-your-fractal

Layer Algebra — a verified algebra of a layered canvas | *mathematical theory; Lean 4 / Mathlib via agentic coding* 2026

- Independently developed the theory of a layered canvas: a canvas maps positions to colours plus transparency; *overdraw* (compositing) makes canvases an idempotent monoid, never a group; a layer pairs a canvas with an orientation; layers quotiented by what they render form a monoid **isomorphic** to the canvases under overdraw. Every result **formalised and machine-checked in Lean 4**.
- Every numbered result proved and verified by machine; reviewed by **Prof. Nathan Carter** (Bentley University, author of *Visual Group Theory*). github.com/rtmkrptn

Discrete Layer Algebra — compositing solver + live demo | *Python, NumPy; JavaScript demo — agentic build* 2026

- Turned the theory into a solver for a compositing puzzle (stack and rotate semi-transparent tiles to match a target image). Trained tabular Q-learning and a DQN, then established that an untrained **greedy best-first search** over a pixel-match measure, with a priority frontier to survive occlusion, solves every difficulty. Live browser demo. github.com/rtmkrptn/discrete-layer-algebra

Smaller work, 2025–2026: XOR network from scratch in pure NumPy (3 layers, backpropagation, gradients by hand); vector-field streamlines, Monte Carlo sampling, random walks, numerical ODE systems; Desmos models of complex-plane paths and Brownian motion. github.com/rtmkrptn/cool-math-things

TECHNICAL SKILLS

Write: Python (NumPy), L^AT_EX

Specify & direct via agentic coding: TypeScript, WebGL2/GLSL, Lean 4 — I architect the system and the hard core (rendering, numerical precision, data model) and implement it through Claude Code

Tools: Git, Claude Code (agentic coding), Desmos

Spoken: Russian (native), English (C1 — IELTS 7.5)

EDUCATION

Byron College

Athens, Greece

Edexcel International Advanced Level — Mathematics, Further Mathematics, Computer Science 2025 – Jun. 2027

- AS results, June 2026:** Mathematics — A (298/300 UMS) • Further Mathematics — A (267/300 UMS) • Computer Science — A. A2 examinations in progress; final results August 2027.

SELF-DIRECTED STUDY

Reading: Herstein, *Topics in Algebra* • Feder, *Fractals* • Stewart, *Calculus* (multivariable). **Courses:** *Probability Theory* (DelftX) • mathematics for machine learning, implemented independently in Python (deep-ml.com) • one-year UX/UI design course.